Jeffrey Hoelzel Jr

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Work Experience

Software Engineer Intern | Altored Health | Flagstaff, AZ **Jan 2026 - Present**

- Architected an LLM-assisted conversational healthcare search system using Node.js APIs and semantic retrieval to map patient queries to care visit targets.
- Built multi-step dialogue workflows that generate follow-up questions to refine patient search intent.

Machine Learning Engineer | Pathogen and Microbiome Institute | Flagstaff, AZ **Aug 2025 - Present**

- Developed PepSeqPred, a PyTorch-based machine learning (ML) pipeline to predict antibody epitope locations across diverse pathogens.
- Optimized ESM-2 embedding generation pipelines using batching and SLURM GPU parallelization, reducing high-performance compute (HPC) runtime by over 50%.
- Built logging and evaluation tools to validate embeddings and model outputs for downstream training and predictions.

Software Engineer, Data Systems | NAU SICCS | Flagstaff, AZ **Jan 2025 - Present**

- Engineered Python-based web scrapers with Selenium and BeautifulSoup to extract unstructured UltraSignUp trail race data for downstream statistical analysis.
- Improved scraping throughput by nearly 90% via concurrency and adaptive wait time strategies.
- Reduced OpenStreetMap API calls by 48.5% through dynamic GPX coordinate batching.

Information Technology Intern | Cavco Industries, Inc. | Phoenix, AZ **Jun 2025 - Aug 2025**

- Built agentic LLM workflows, enabling automated ServiceDesk Plus ticket operations.
- Designed a retrieval-augmented generation (RAG) pipeline, reducing response latency by 60% and consolidating 3+ API calls into 1 request.
- Developed an ML pipeline clustering 100,000+ tickets to automatically generate Confluence knowledge articles.

Software Engineer, Applied AI | CANIS Lab | Flagstaff, AZ **Jan 2025 - Jul 2025**

- Built a campus health chatbot using ReactJS and Flask to support student health inquiries and facilitate research studies.
- Evaluated multiple language models on multi-GPU cloud compute systems, enabling sub-second response times.
- Improved response relevance by 35% by refining RAG pipelines and contextual ranking.

Projects

ArtemiS3 | Capstone Project (USGS and NASA) **Svelte | FastAPI | PostgreSQL | Boto3 | Meilisearch**

- Architected a Dockerized search platform enabling discovery across public NASA and USGS AWS S3 datasets.
- Built REST API endpoints with FastAPI and optimized queries using Meilisearch and Boto3.
- Led technical planning and stakeholder presentations, delivering architecture, feasibility, and requirements documentation.

MACH-IV Clustering | Unsupervised Machine Learning (CS472) **Scikit-Learn | SciPy | Pandas | NumPy**

- Designed and implemented an end-to-end clustering pipeline analyzing 70,000+ participant responses.
- Applied k-means, Gaussian mixture, hierarchical, and spectral clustering to identify latent behavioral patterns.
- Architected the project codebase and deployed an interactive results site for model exploration.

Louie's Ratings | Software Engineering (CS386) **ReactJS | Flask | PostgreSQL**

- Led a five-person team through sprint planning and delivery of a full-stack academic rating and grade insights platform.
- Developed secure authentication flows using ReactJS and Flask with bcrypt password hashing.
- Implemented unit and end-to-end acceptance testing with unittest and Selenium to ensure deployment stability.

Education

Northern Arizona University, Flagstaff, AZ **GPA: 3.85**

BSc in Software Engineering | Minor in Mathematics *Lumberjack Scholar | HURA Grant Recipient | Dean's List*

Technical Skills

Languages: Python, JavaScript, TypeScript, C, Java, SQL, HTML, CSS

Libraries/Frameworks: ReactJS, Svelte, Flask, FastAPI, PyTorch, Scikit-Learn, Pandas, SQLite, PostgreSQL, Firebase Firestore

Developer Tools: Docker, Git, Slurm (Bash), AWS, Google Cloud, HPC, CI/CD, Agile